Problem Set 3

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For every single question, unless otherwise noted I collaborated with Blair Moore, Thomas Li, and John Cho.

Problem 1

Problem 1.a

The conditional log-likelihood doesn't depend on the distribution of x. So we only care about the probability of a given observation given x. Namely the likelihood a given observation i observation being y_i subject to parameters estimates β and covariates x_i is:

$$P(y_i|x_i) = \frac{(e^{x_i'\beta})^{y_i} e^{-e^{x_i'\beta}}}{y_i!}$$

Which is a bit of mess. But we can simplify it when we take the log:

$$\begin{aligned}
\ell(z;\beta) &= \log\left(\prod_{i\in N} \frac{(e^{x_i'\beta})^{y_i} e^{-e^{x_i'\beta}}}{y_i!}\right) \\
&= \sum_{i\in N} \log\left(\frac{(e^{x_i'\beta})^{y_i} e^{-e^{x_i'\beta}}}{y_i!}\right) \\
&= \sum_{i\in N} \log((e^{x_i'\beta})^{y_i}) + \sum_{i\in N} \log(e^{-e^{x_i'\beta}}) - \sum_{i\in N} \log(y_i!) \\
&= \sum_{i\in N} y_i \log(e^{x_i'\beta}) - \sum_{i\in N} e^{x_i'\beta} - \sum_{i\in N} \log(y_i!) \\
&= \sum_{i\in N} y_i x_i'\beta - \sum_{i\in N} e^{x_i'\beta} - \sum_{i\in N} \log(y_i!) \\
&= \sum_{i\in N} y_i x_i'\beta - \sum_{i\in N} e^{x_i'\beta} - \sum_{i\in N} \underbrace{\sum_{k=1}^{y_i} \log(k)}_{\text{Sometimes empty}} \\
&= \sum_{i\in N} y_i x_i'\beta - \sum_{i\in N} e^{x_i'\beta} - \sum_{i\in N} \sum_{k=1}^{y_i-1} \log(k+1)
\end{aligned}$$

Where the last two lines are alternate forms of the same thing (I mean duh, that's all of algebra) but with different ways to index the log of the factorial. That's really more of a curiosity for me, because those won't affect the maximizing choice of β anyway, but it's kind of nice that the conditional log-likelihood function depends on something like the "total" number of events across all people (where each person might have multiple events according to the Poisson distribution).

Problem 1.b

The score is the derivative of an individual's log-likelihood with respect to the estimands, in our case β. In our case, that would be:

$$\begin{aligned} s_\beta(z_i) &= \frac{\partial}{\partial \beta}\left[y_i x_i'\beta - e^{x_i'\beta} - \log(y_i!)\right] \\ &= y_i x_i - e^{x_i'\beta} x_i \\ &= x_i(y_i - e^{x_i'\beta}) \end{aligned}$$

Which despite its simple form, I'm pretty sure that's a transcendental equation that we have no hope of solving in closed form for one person, let alone across the population.

And just in case you were looking for the population score function, it would be this:

$$\sum_i x_i(y_i - e^{x_i'\beta})$$

To compute the Hessian H matrix, we need to take a second derivative of the score function with respect to β' giving:

$$\begin{aligned} H &= \frac{\partial}{\partial \beta'} E\left[x(y - e^{x'\beta})\right] \\ &= E\left[\frac{\partial}{\partial \beta'} x(y - e^{x'\beta})\right] \\ &= E\left[\frac{\partial}{\partial \beta'} xy - \frac{\partial}{\partial \beta'} x e^{x'\beta}\right] \\ &= E\left[0 - \frac{\partial}{\partial \beta'} x e^{x_i'\beta}\right] \\ &= -E\left[x \frac{\partial}{\partial \beta'} e^{x'\beta}\right] \\ &= -E\left[x e^{x'\beta} x'\right] \\ &= -E\left[e^{x_i'\beta} x x'\right] \end{aligned}$$

And without knowing more about the distribution of x, that's the best we can do.

The J matrix is given by:

$$\begin{aligned} J &= E[s_\beta(z) s_\beta(z)'] \\ &= E\left[\left(x_i(y_i - e^{x_i'\beta})\right)\left(x_i(y_i - e^{x_i'\beta})\right)'\right] \\ &= E\left[(y_i - e^{x_i'\beta})^2 x_i x_i'\right] \end{aligned}$$

Problem 1.c

We define

$$\beta_0 \equiv \arg\min_\beta Q_{\beta_0}(\beta)$$

Where $Q(\beta) = E_{\beta_0}\left[\log(f_{\beta_0}(y|x)) - \log(f_\beta(y|x))\right]$ We'd like to show that the in sample, $\hat{\beta}$ converges in probability to that same β_0 as $n \to \infty$ so long as $E[y|x] = \exp(x'\beta)$. Since β_0 is the minimizer of a function, it is natural to employ the results from M-estimators that we learned in section.

In particular, to show that that the in sample estimator of the minimizer, $\hat{\beta}$ is consistent for β_0, we must establish four conditions:

1. $E_{\beta_0}\left[\ell(z;\beta)\right]$ is uniquely minimized at β_0
2. The space of possible βs is compact
3. $E_{\beta_0}\left[\ell(z;\beta)\right]$ is continuous
4. The sample analog of $E_{\beta_0}\left[\ell(z;\beta)\right]$ converges uniformly in probability to $E_{\beta_0}\left[\ell(z;\beta)\right]$

Then, we have learned that this directly implies that $\hat{\beta}$ is consistent for β_0.

Let's tackle each of those four separately. For 1., note that the hessian of Q would be the same as the hessian derived above because the only parts that depend on β are the $f_\beta(y|x)$, which is exactly the conditional likelihood derived above. Then note that $-E\left[e^{x_i'\beta}xx'\right]$ is negative semi-definite (the suff on the inside is positive or an outer product which is psd), so the function is globally concave, so there is a unique maximizer. In the population, the true β_0 must satisfy the FOC:

$$
\begin{aligned}
0 &= \frac{\partial}{\partial\beta} E_{\beta_0}\left[\log(f_{\beta_0}(y|x)) - \log(f_\beta(y|x))\right] \\
&= E_{\beta_0}\left[\frac{\partial}{\partial\beta}\log(f_{\beta_0}(y|x)) - \frac{\partial}{\partial\beta}\log(f_\beta(y|x))\right] \\
&= E_{\beta_0}\left[0 - \frac{\partial}{\partial\beta}\log(f_\beta(y|x))\right] \\
&= E_{\beta_0}\left[-\frac{\partial}{\partial\beta}\log(f_\beta(y|x))\right] \\
&= E_{\beta_0}\left[-\frac{\partial}{\partial\beta}(yx'\beta - \exp(x'\beta))\right] \\
&= E_{\beta_0}\left[-(yx - \exp(x'\beta)x)\right] \\
&= E_{\beta_0}\left[-(y - \exp(x'\beta))x\right] \\
&= E_{\beta_0}\left[E_{y|x}[-(y - \exp(x'\beta))x|x]\right] \\
&= E_{\beta_0}\left[-(E_{y|x}[y] - \exp(x'\beta))x|x]\right] \\
&= E_{\beta_0}\left[-(\exp(x'\beta_0) - \exp(x'\beta))x|x]\right]
\end{aligned}
$$

Which clearly requires setting β to be β_0. The last line comes from the assumption that the true conditional mean is as expected, even if the distribution is wrong.

So $\beta = \beta_0$ is the unique maximizer

For 2., here we would use economic intution to say put reasonable bounds on the values that β_0 might take in reality and only consider the compact set between those values.

For 3., the function is clearly continuous.

For 4. For uniform convergence of $\hat{Q}$ to Q, we need to show that for all true values of β_0, $\sup_\beta(|\hat{Q}(\beta) - Q(\beta)|) \to^p 0$, that is no values of β stay extremely far away. This would require showing that the maximum of the following expression:

$$
\sum_{i\in N} y_i x_i'\beta - \sum_{i\in N} e^{x_i'\beta} - \sum_{i\in N}\sum_{k=1}^{y_i-1}\log(k+1)
$$

which defines $\hat{\beta}$ goes to zero, no matter the value of β.

This would complete the proof

Problem 2

Problem 2.a

The likelihood for a given y_i would be:

$$\ell(y|x) = \log\left(\Phi(x'\beta)^y \cdot (1-\Phi(x'\beta))^{1-y}\right) = y\log(\Phi(x'\beta)) + (1-y)\log(1-\Phi(x'\beta))$$

And the score is the derivative with respect to β, so we have:

$$\begin{aligned}
s_\beta &= \frac{\partial}{\partial\beta}\left[y\log(\Phi(x'\beta)) + (1-y)\log(1-\Phi(x'\beta))\right] \\
&= y\frac{\partial}{\partial\beta}\log(x'\beta) + (1-y)\frac{\partial}{\partial\beta}\log(1-\Phi(x'\beta)) \\
&= y\frac{\phi(x'\beta)}{\Phi(x'\beta)}x + (1-y)\frac{-\phi(x'\beta)}{1-\Phi(x'\beta)}x \\
&= x\left(y\frac{\phi(x'\beta)}{\Phi(x'\beta)} + (1-y)\frac{-\phi(x'\beta)}{1-\Phi(x'\beta)}\right) \\
&= x\frac{(y-\Phi(x'\beta))\phi(x'\beta)}{\Phi(x'\beta)(1-\Phi(x'\beta))} \\
&= x(y-\Phi(x'\beta))\frac{\phi(x'\beta)}{\Phi(x'\beta)(1-\Phi(x'\beta))} \\
&\equiv x(y-\Phi(x'\beta))\alpha(x'\beta)
\end{aligned}$$

Which probably has a really nice interpretation in terms of hazard rates because in either binary case of y, the score cancels into something that looks exactly like a hazard rate. The problem is that the α function is virtually intractable to work with (or I'm bad at calculus), so I will never fully characterize $\alpha'(x'\beta)$. because it basically doesn't simplify.

The Hessian is then

$$\begin{aligned}
H &= \frac{\partial}{\partial\beta'}E\left[x(y-\Phi(x'\beta))\alpha(x'\beta)\right] \\
&= E\left[\frac{\partial}{\partial\beta'}x(y-\Phi(x'\beta))\alpha(x'\beta)\right] \\
&= E\left[x\left(-\phi(x'\beta)x'\right)\alpha(x'\beta) + (y-\Phi(x'\beta))\alpha'(x'\beta)x')\right] \\
&= E\left[xx'\left(-\phi(x'\beta)\right)\alpha(x'\beta) + (y-\Phi(x'\beta))\alpha'(x'\beta))\right] \\
&= E\left[E[xx'\left(-\phi(x'\beta)\right)\alpha(x'\beta) + (y-\Phi(x'\beta))\alpha'(x'\beta))\,|x]\right] \\
&= E\left[xx'\left(-\phi(x'\beta)\right)\alpha(x'\beta) + E[y-\Phi(x'\beta)|x]\alpha'(x'\beta))\right] \\
&= -E\left[xx'\phi(x'\beta)\alpha(x'\beta)\right] + E\left[xx'E[y-\Phi(x'\beta)|x]\alpha'(x'\beta))\right] \\
&= -E\left[xx'\phi(x'\beta)\frac{\phi(x'\beta)}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right] + E\left[xx'(E[y|x]-\Phi(x'\beta))\alpha'(x'\beta))\right] \\
&= -E\left[xx'\frac{\phi(x'\beta)^2}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right] + E\left[xx'(E[y|x]-\Phi(x'\beta))\alpha'(x'\beta))\right]
\end{aligned}$$

Gnarly, but it will be useful to have decomposed into a structural component, and then one that depends on the possible misspecification.

Anyway, let's compute the J matrix:

$$
\begin{aligned}
J &= E[s_\beta \cdot s_\beta'] \\
&= E\left[(x(y-\Phi(x'\beta))\alpha(x'\beta))\,(x(y-\Phi(x'\beta))\alpha(x'\beta))'\right] \\
&= E\left[xx'\alpha(x'\beta)^2\,(y-\Phi(x'\beta))^2\right] \\
&= E\left[xx'\alpha(x'\beta)^2 E\left[(y-\Phi(x'\beta))^2\middle|x\right]\right] \\
&= E\left[xx'\left(\frac{\phi(x'\beta)}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right)^2 E\left[(y-\Phi(x'\beta))^2\middle|x\right]\right] \\
&= E\left[xx'\frac{\phi(x'\beta)^2}{\Phi(x'\beta)^2(1-\Phi(x'\beta))^2} E\left[(y-\Phi(x'\beta))^2\middle|x\right]\right]
\end{aligned}
$$

Problem 2.b

Correct model specification means that $Pr(y=1|x) = \Phi(x'\beta)$. Noting that the expectation of a binary rv is simply $E[y|x] = Pr(y=1|x)\cdot 1$, so in our case, the expected value of $E[y|x]$ is simply $\Phi(x'\beta)$. That means that $E[(y-\Phi(x'\beta))^2|x]$ is actually just the variance of the bernoulli random variable since we're considering squared deviations of the binary y from its mean. Its variance would then be $\Phi(x'\beta)(1-\Phi(x'\beta))$.

Recalling from the previous question:

$$
\begin{aligned}
H &= -E\left[xx'\frac{\phi(x'\beta)^2}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right] + E\left[xx'(E[y|x]-\Phi(x'\beta))\alpha'(x'\beta))\right] \\
&= -E\left[xx'\frac{\phi(x'\beta)^2}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right] + E\left[xx'(\Phi(x'\beta)-\Phi(x'\beta))\alpha'(x'\beta))\right] \\
&= -E\left[xx'\frac{\phi(x'\beta)^2}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right] + E\left[xx'(0)\alpha'(x'\beta))\right] \\
&= -E\left[xx'\frac{\phi(x'\beta)^2}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right] \\
J &= E\left[xx'E\left[(y-\Phi(x'\beta))^2|x\right]\frac{\phi(x'\beta)^2}{\Phi(x'\beta)^2(1-\Phi(x'\beta))^2}\right] \\
&= E\left[xx'\Phi(x'\beta)(1-\Phi(x'\beta))\frac{\phi(x'\beta)^2}{\Phi(x'\beta)^2(1-\Phi(x'\beta))^2}\right] \\
&= E\left[xx'\frac{\phi(x'\beta)^2}{\Phi(x'\beta)(1-\Phi(x'\beta))}\right]
\end{aligned}
$$

Coolio, that verifies the identity.

Problem 2.c

I already found the conditional log-likelihood for an individual observation. So the conditional log likelihood for n data points would be:

$$
\sum_i \ell(y_i|x_i) = y_i\log(\Phi(x_i'\beta)) + (1-y_i)\log(1-\Phi(x_i'\beta))
$$

Problem 2.d

Unfortunately, this question will require me to compute the derivative of my α function, though I won't attempt to simplify it. See the following:

$$
\begin{aligned}
\frac{d}{dz}\alpha(z) &= \frac{d}{dz}\left[\frac{\phi(z)}{\Phi(z)(1-\Phi(z))}\right] \\
&= \frac{\phi'(z)\Phi(z)(1-\Phi(z)) - \phi(z)(\phi(z)(1-\Phi(z)) - \Phi(z)\phi(z))}{\Phi(z)^2(1-\Phi(z))^2}
\end{aligned}
$$

And regrettably, while `R` has implemented ϕ and Φ, it doesn't have a default ϕ' implementation. So we must compute it explicitly:

$$
\begin{aligned}
\phi'(z) &= \frac{d}{dz}\frac{1}{\sqrt{2\pi}}e^{-z^2/2} \\
&= \frac{-z}{\sqrt{2\pi}}e^{-z^2/2}
\end{aligned}
$$

First we implement the formulae used to compute the H and J matrix for a given x and y.

```
# test values
x <- matrix(c(1, 1, 32, 11), ncol = 1)
y <- 1
beta <- matrix(c(2.08022, -0.79950, -0.03482, -0.01145), ncol = 1)

alpha_prime <- function(z){
  Phi <- pnorm(z)
  phi <- dnorm(z)
  phi_prime <- -z/sqrt(2*pi)*exp(-(z^2)/2)

  numerator <-
    (phi_prime * Phi * (1-Phi)) -
    (phi * ((phi * (1-Phi)) - (phi * Phi)))
  denominator <- (Phi^2) * ((1-Phi)^2)

  numerator / denominator
}

H <- function(x,y,beta){
  stopifnot(all(dim(x)==dim(beta)))
  stopifnot(length(y)==1)

  x_prime_beta <- {t(x) %*% beta}[1,1]
  phi <- dnorm(x_prime_beta)
  Phi <- pnorm(x_prime_beta)


  first_exp = x %*% t(x) * (phi^2) / (Phi*(1-Phi))
  second_exp = x %*% t(x) * (y - Phi) * alpha_prime(x_prime_beta)

  -first_exp + second_exp
}

J <- function(x,y,beta,correctly_specified = F){
  stopifnot(all(dim(x)==dim(beta)))
  stopifnot(length(y)==1)

  x_prime_beta <- {t(x) %*% beta}[1,1]
```

```r
  phi <- dnorm(x_prime_beta)
  Phi <- pnorm(x_prime_beta)

  if (correctly_specified){
    inner_exp <- (Phi)*(1-Phi)
  } else{
    inner_exp <- (y-Phi)^2
  }
  frac <- (phi/(Phi*(1-Phi)))^2

  x %*% t(x) * frac * inner_exp
}

# H(x, y, beta)
# J(x, y, beta)
# J(x, y, beta, correctly_specified = T)
```

Now we create functions to estimate standard errors from a dataset.

```r
estimate_hat <- function(X, y, beta, estimator, ...) {

  # Convert each row of X into a named column vector (matrix ncol = 1)
  X_list <- apply(
    X, 1,
    function(m) {
      mat <- matrix(m, ncol = 1)
      rownames(mat) <- names(m)
      mat
    },
    simplify = FALSE
  )

  # Evaluate estimator on each (x_i, y_i), forwarding extra args
  hats <- mapply(
    function(x_i, y_i) estimator(x_i, y_i, beta, ...),
    X_list, y,
    SIMPLIFY = FALSE
  )

  # Average (sum then divide)
  Reduce("+", hats) / length(hats)
}
```

Now we load the data and actually compute standard errors.

```r
# explicitly make a column vector
beta_hat <- matrix(c(2.08022, -0.79950, -0.03482, -0.01145), ncol = 1)

data <- read.csv('mroz.csv')
y <- data$part
# one row per data point. effectively when we slice out a row, we already
# get an x' (x transpose)
X <- cbind(rep(1,nrow(data)), as.matrix(data[,c('kidslt6', 'age', 'nwifeinc')]))
colnames(X)[1] <- "(Intercept)"
```

```r
H_hat <- estimate_hat(X, y, beta_hat, H)
J_hat <- estimate_hat(X, y, beta_hat, J)
J_hat_correctly_specified <- estimate_hat(X, y, beta_hat, J,
                                          correctly_specified = T)

sandwich <- solve(H_hat)%*%J_hat%*%solve(H_hat)
estimated_variance <- sandwich/length(y)
estimated_ses <- sqrt(diag(estimated_variance))

estimated_variance_correctly_specified <- solve(J_hat)/length(y)
estimated_ses_correctly_specified <- sqrt(diag(estimated_variance_correctly_specified))

estimated_variance_alt <- solve(J_hat_correctly_specified)/length(y)
estimated_ses_alt <- sqrt(diag(estimated_variance_alt))
```

```
## Sandwich QMLE SEs:
## (Intercept)     kidslt6         age    nwifeinc
## 0.307638462 0.112448536 0.006634194 0.004333156
```

```
## Correct Specification SEs:
## (Intercept)     kidslt6         age    nwifeinc
## 0.310395956 0.107105005 0.006674365 0.003969286
```

```
## Completely Correct Specification SEs:
## (Intercept)     kidslt6         age    nwifeinc
## 0.308675005 0.110531541 0.006649356 0.004191699
```

Problem 3

Problem 3.a

The data available is

$$
\begin{aligned}
y_{A1} &= \alpha_A \cdot 1 + \alpha_B \cdot 0 + \beta_1 \cdot 1 + \beta_2 \cdot 0 + \tau \cdot 0 + \varepsilon \\
y_{A2} &= \alpha_A \cdot 1 + \alpha_B \cdot 0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \tau \cdot 1 + \varepsilon \\
y_{B1} &= \alpha_A \cdot 0 + \alpha_B \cdot 1 + \beta_1 \cdot 0 + \beta_2 \cdot 0 + \tau \cdot 0 + \varepsilon \\
y_{B2} &= \alpha_A \cdot 0 + \alpha_B \cdot 1 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \tau \cdot 0 + \varepsilon
\end{aligned}
$$

$$
\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}
\begin{bmatrix} \alpha_A \\ \alpha_B \\ \beta_1 \\ \beta_2 \\ \tau \end{bmatrix}
$$

Clearly $X'X$ will not be invertible since we're trying to estimate more parameters than rows in the data. So with the data we have, we had no hope of identifying the αs separately from the βs

In particular, column 1 plus column 2 minus column 3 equals column 4. So the fixed effects can't be estimated.

But the problem is actually deeper than that. Identification requires that different values of the parameters produce different distributions of the data. I'll argue that two different sets of fixed effects parameters produce identical distributions of the data.

Suppose that all of the individual fixed effects were larger by an additive factor η and all of the time fixed effects were smaller by that same factor. Then every observation has an individual and a time period, they

experience a $+\eta$ and a $-\eta$ which produces a net zero effect on that observation. This was actually a generic argument that doesn't require the 2-person 2-period setup we have here. This proves generally that they can't be jointly identified. Notice that if we force $\beta_1 = 0$, then there would be observation that would have a $-\eta$ but not a $+\eta$, so different values of the parameters produce different data.

Problem 3.b

When $\beta_1 = 0$, we can rewrite the above data/matrix as:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha_A \\ \alpha_B \\ \beta_2 \\ \tau \end{bmatrix}$$

Desmos tells me:

$$(X'X)^{-1}X' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}$$

So for an arbitrary $[y_{A1}\ y_{A2}\ y_{B1}\ y_{B2}]'$, the $\hat{\tau}$ is recovered from the inner product with the final row:

$$[-1\ 1\ 1\ -1][y_{A1}\ y_{A2}\ y_{B1}\ y_{B2}]' = -y_{A1} + y_{A2} + y_{B1} - y_{B2} = (y_{A2} - y_{A1}) - (y_{B2} - y_{B1})$$

Which is exactly the DiD estimator.